

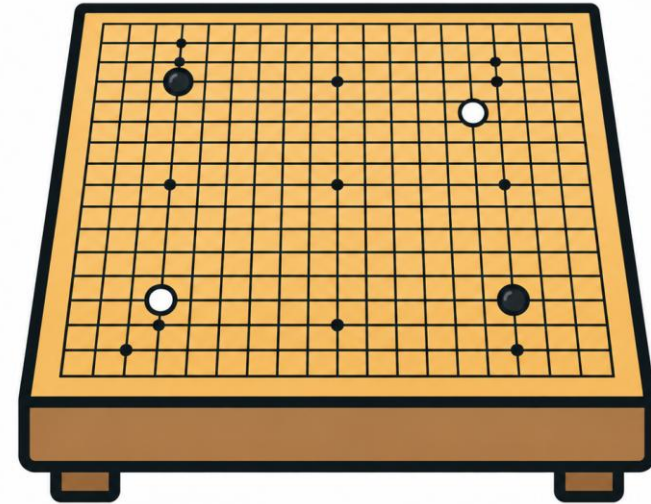
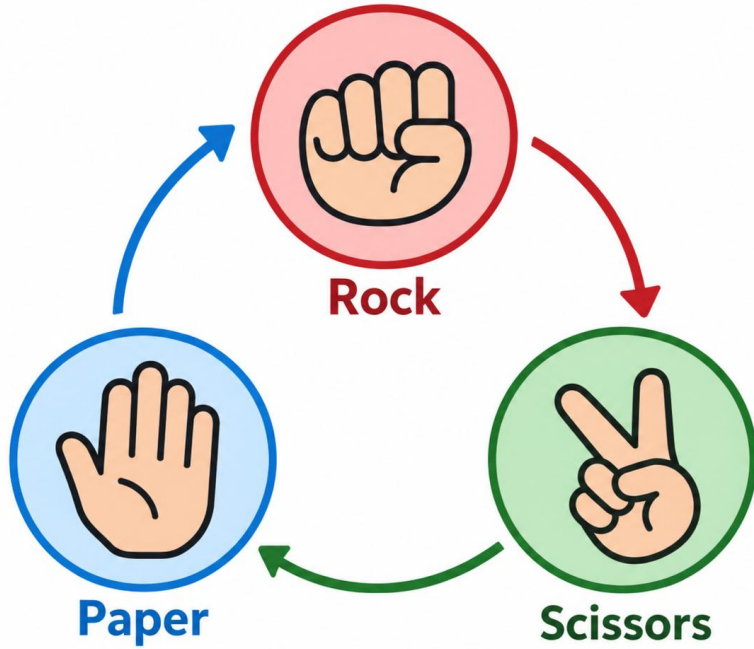
Two-Player Zero-Sum Games

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References

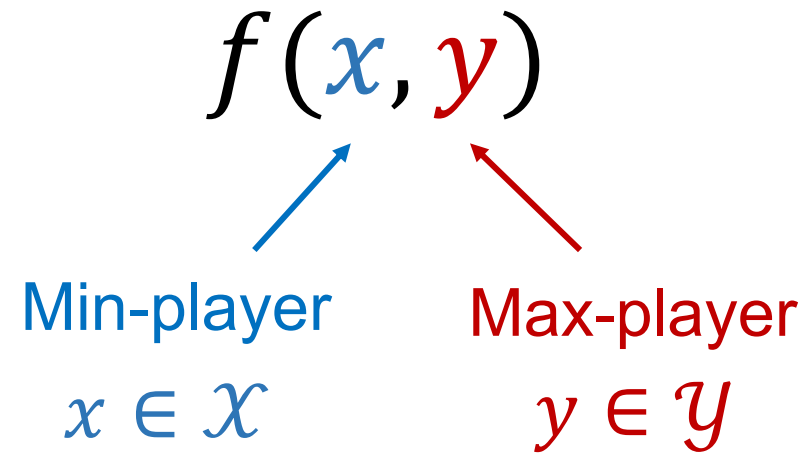
- Lecture 4 of [Haipeng's course](#)

Two-Player Zero-Sum Games



Go

Two-Player Zero-Sum Games



$$\mathcal{X} = \{ \text{Rock, paper, scissor} \}$$

$$\mathcal{X} = \Delta_3$$

Game Matrix

$$G = \begin{bmatrix} 1 & 1 & 0 & -3 \\ 2 & 3 & 1 & -5 \end{bmatrix}$$

cost matrix for row-player

Column player

			
 row player	0	0.99 -1	-1
	-1	0	1
	1	-1	0

$\Rightarrow 0.99$

$\Rightarrow 1$

$\Rightarrow 1$

$$\sum_{i=1}^3 \sum_{j=1}^3 p(i) g(j) G(i,j)$$

$$= p^T G g$$

$$p \in \Delta_N, g \in \Delta_M$$

$$G \in \mathbb{R}^{N \times M}$$

If row-player uses $p \in \Delta_3$, column player uses $g \in \Delta_3$

Discussion (Sep 16)

Given a game matrix, (assuming the opponent will do their best), what is the best we can do?

Compare the Two Values

If we (min-player) have to play under this protocol, then by choosing

$$x^* = \operatorname{argmin}_x g(x) = \operatorname{argmin}_x \left(\max_y f(x, y) \right)$$

my cost is at most

$$\min_x \max_y f(x, y)$$

Proof,

$$\text{my cost} = f(x^*, \hat{y})$$

$$\leq \max_y f(x^*, y)$$

$$= \min_x \max_y f(x, y)$$

Max has to tell min their decision

their value $\geq \max_y \min_x f(x, y)$

$$\min_x \left(\max_y f(x, y) \right) \geq \max_y \left(\min_x f(x, y) \right)$$

$\uparrow$ min $\uparrow$ max
 $\downarrow$ depend on f $\downarrow$ h(y)

$g(x) = \text{my cost if I choose } x$

→ Protocol: min-player has to tell max-player their decision

min $\hat{x}$ max $\hat{y}$

Compare the Two Values

$$\begin{aligned} \text{let } x^* &= \operatorname{argmin}_x \left(\max_y f(x, y) \right) \\ \min_{x \in X} \left(\max_{y \in Y} f(x, y) \right) &= \max_{y \in Y} \left(f(x^*, y) \right) \\ &\geq \max_{y \in Y} \left(\min_{x \in X} f(x, y) \right) \end{aligned}$$

<https://www.youtube.com/watch?v=RmgLPeASjXs>

Minimax Theorem Informal Illustration
(but this is actually not the minimax theorem...)

Compare the Two Values

Game values when both players do their best:

$$\min_x \max_y f(x, y)$$

Protocol: min-player has to reveal their decision (x) to max-player

$$\max_y \min_x f(x, y)$$

Protocol: max-player has to reveal their decision (y) to min-player

Minimax Theorem

(John von Neumann's) Minimax Theorem (1928).

Let $G \in \mathbb{R}^{N \times M}$. Then

$$\min_{p \in \Delta_N} \max_{q \in \Delta_M} p^\top G q = \max_{q \in \Delta_M} \min_{p \in \Delta_N} p^\top G q$$

Discussion

Given a game matrix, (assuming the opponent will do their best), what is the best we can do?

Proof by running algorithm

Nash Equilibrium

Another Proof of the Minimax Theorem

Application: Boosting

Setting: Binary classification

Question: Suppose we can always find a classifier that is only slightly better than random (i.e., 50% accuracy). Can we use this **weak learner** to produce an accurate classifier?

Setting

Training data: $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N) \in \mathcal{F} \times \{-1, 1\}$

Classifier: $h: \mathcal{F} \rightarrow \{-1, 1\}$

Loss of a classifier on a training sample:

What Does “Weak Learner” Mean?

Applying Minimax Theorem

From No-Regret Learning to Boosting

Input: Training dataset $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$

Input: Classification oracle $\mathcal{O}$: weighted dataset $\rightarrow$ classification model

Initialize a no-regret algorithm $\mathcal{A}$ over Δ_N

For $t = 1, \dots, T$:

Let no-regret algorithm $\mathcal{A}$ output $p_t \in \Delta_N$ (distribution over data samples)

Let oracle $\mathcal{O}$ output a classification model h_t :

$$h_t = \operatorname{argmin}_{h \in \mathcal{H}} \mathbb{E}_{i \sim p_t} [1[h(x_i) \neq y_i]]$$

Define **reward** $r_t(i) = 1[h_t(x_i) \neq y_i]$.

Feed reward vector r_t to no-regret algorithm $\mathcal{A}$.

Caveat

- The discussion is only about the “training error”